

# TIN DANG HUYNH TRUNG

## Senior AI/LLM Engineer – Python, Retrieval & Production LLM Systems

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I'm a backend engineer, mostly Python, eight years in. I've had machine learning running in production since 2019, and for the last two years I've worked on the infrastructure under LLM products: multi-provider gateways, routing and failover, retrieval, agent memory, MCP, and metering that reconciles against what providers actually bill. The thing I keep coming back to is evaluation — shipping an LLM feature that looks right is easy, knowing it's still right three changes later is not, so I put that in CI rather than checking by hand before a release. I started in fintech and I still build like it: append-only ledgers, tenancy that fails closed, money that can't move twice. An agent writes most of my code now; deciding whether it's correct is still my job, and I published the method I use for that.

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer / Technical Lead

#### Trustify Technology

Jan 2024 – Present

Ho Chi Minh City, Vietnam

I architect and lead client rebuilds across Vietnam and Myanmar — fintech, real estate and intelligent transport for Yoma Group, Yoma Bank, Wave Money and TruStone — and wrote the company's engineering career ladder. I also built Yoma Bank's microfinance lending MVP in Java 21 and Spring Boot, and owned money and authentication on Star City's resident platform for a team of nine.

- **Estate-finance platform & PCI boundary** • *Yoma Group, Wave Money — sole engineer & architect*

I rebuilt a live property-finance system off PHP/Laravel onto a single append-only double-entry ledger where every balance can be re-derived and audited, graded 235 legacy business rules against it before anyone moved, and shipped it in eight releases over four weeks. Cardholder data sits behind its own service, database and network, so only that service falls inside PCI scope and the build rejects any import that crosses the boundary.

Python 3.12 • FastAPI • async SQLAlchemy 2 • PostgreSQL (schema-per-tenant) • Redis • Docker • 2C2P • AWS ECS • Envoy

- **Real-time ML serving at the edge** • *intelligent transport — architect & sole engineer*

400 camera streams produce more video than a city network can carry, so I moved inference to the edge and sent only events onward: YOLO on TensorRT and Triton at a P95 of 85 ms, 75% less backhaul, events reaching the data centre over mutually authenticated Kafka, load-tested to 71K events a second. On a separate device-management migration, rehearsing the rollback is what found the gap — Postgres reverted cleanly, the document store, cache and archived objects did not.

Python • FastAPI • Kafka • PyFlink • TensorRT • Triton • K3s • ArgoCD • DuckDB • PySpark • Airflow

### CTO – Co-founder

#### ClickAI Viet Nam

Jun 2024 – Oct 2025 (concurrent)

Ho Chi Minh City, Vietnam

I co-founded ClickAI (clickai.vn) and built its conversational AI platform from scratch — multi-modal chat, image and audio generation, community — to 40K+ users across Southeast Asia inside four months. As CTO I set architecture and engineering standards, led twelve to fourteen people, and owned billing, identity and deployment myself.

- **Model platform** • *20+ providers*

I built the orchestration layer that routes across 20 third-party providers and one of our own, with failover, vector memory and reasoning-model support; choosing the model by policy rather than habit cut API spend by 20%. The first time a provider went hard-down, users hit the failure before routing shifted — so I stopped waiting for failures and probed providers continuously, marking one down before traffic could reach it. Image and audio generation fans out behind circuit breakers.

Python • FastAPI • Weaviate • Qdrant • FalkorDB • ComfyUI • Replicate • fal.ai • ElevenLabs

- **Billing & metering** • *sole author*

I built the wallet, plan, promotion and referral-commission engine over a month-partitioned ledger that meters every LLM call. A retry loop was quietly re-sending the same work, and a reconciliation check I wrote is what caught it — provider billing had drifted more than 30% from what we metered. I corrected the ledger model and its indexes, keyed idempotency on the source event so a retry cannot double-count, and made the check a standing job.

Python • FastAPI • PostgreSQL (partitioned) • Redis

- **Production, identity & web** • *99.9% uptime, 1K+ concurrent sessions*

I authored the 13-service deployment with cert-manager and autoscaling, and held sub-2s responses. When the database saturated I moved PostgreSQL onto a three-node Patroni cluster; when a release regressed, my own instrumentation caught it before any customer did; and a hosting comparison I took to stakeholders halved the bill.

Kubernetes • Helm • Azure • Patroni • Prometheus • Next.js 14 • TypeScript

### Founder & Principal Engineer

#### Pilot Space (open-source AI infrastructure)

Jan 2026 – Present

Remote

- **Hydroa** • *multi-tenant LLM gateway*  
One OpenAI-compatible endpoint over OpenAI, Anthropic, Gemini, Bedrock, Azure and OpenRouter, metering what each tenant spends. Envoy authorises at the edge, providers sit behind per-tenant and per-provider circuit breakers, customer keys are encrypted at rest, and a cost ledger reconciles against what was actually billed. Tests outweigh source two to one by line.  
Python · FastAPI · Envoy · PostgreSQL · Kubernetes · OpenTelemetry
- **Lunaris & Helios** • *retrieval pipeline and agent harness*  
Lunaris is ingest, extraction, embeddings, vector search and reranking behind one API, recalling across 100K documents per scope in 19–22 ms at the median. Its quality is gated rather than asserted: CI scores LongMemEval-S evidence recall judge-free, on both operating points, and fails the build when recall drops past tolerance or the retrieval config moves under the baseline. I wrote that gate after finding its predecessor had never once run — 200 of 200 recorded executions were startup failures, so nothing had been gated at all. Helios consumes it: 12 provider backends, a native MCP client and server, and a circuit breaker on every outbound call.  
Rust · Tokio · llama.cpp embeddings · MCP · Wasmtime · npm / PyPI / crates.io
- **Moon** • *Redis-compatible in-memory store, written from scratch*  
Existing Redis clients connect unchanged: RESP2/RESP3, 250+ commands, and native vector, BM25 and graph search under MVCC. Thirty releases to v0.8.9, measuring 1.62× on GET and 1.32× on SET against Redis 7.0 with 8–22% less memory per key — published alongside the command families where it loses.  
Rust · Tokio · io\_uring · crates.io · PyPI
- **ADD** • *the method behind the projects above*  
An agent writes most of my code, so I made the trust explicit: the contract is frozen and hashed before any code exists, the failing test is committed before the fix, and a build is accepted on a receipt binding the passing run to the files that produced it — never on a diff that reads well. I published a six-milestone benchmark of the method, then retracted my own headline claim after finding a defect in my own meter; the corrected report has it costing 1.7× the cheaper alternative on a workload that could not tell them apart. Shipped as @pilotspace/add and pilotspace-add at 3.6.0.

## EARLIER EXPERIENCE (2018 – 2024)

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- **Senior Data Engineer**, Asia Commercial Bank (ACB) • Aug 2023 – Jun 2024  
Ran an enterprise data lake carrying 5M+ daily transactions. A data-quality framework I built across six dimensions cut manual verification by 80%, and my APIs served 100+ internal applications. Cloudera · PySpark · Dagster · Iceberg · FastAPI
- **Software Data Engineer**, Vietmoney Corp • Mar 2021 – Aug 2023  
Built on a fintech appraisal and lending platform running 10M+ daily transactions at 99.9% uptime. Price forecasting at 96% accuracy cut appraisal from 15 minutes to 5 seconds, and a distributed crawler doubled throughput across 50+ SOURCES. Python · PyTorch · BERT · Scrapy · Pub/Sub · BigQuery · Airflow · GCP
- **Software Developer**, Dai Phat Solutions • Aug 2019 – Mar 2021  
Shipped CCTV analytics at 15 FPS to 20+ clients, cutting reported incidents by 40%, and face recognition at 92% accuracy over 1,000+ employees that removed 90% of a manual process. Python · PyTorch · OpenCV · FAISS · Kafka · FFmpeg
- **Full-stack Developer**, LogiGear Corporation • Jun 2018 – Mar 2019  
Built Ethereum wallet, smart-contract and liquidity-pool management in Node.js and React.

## OPEN SOURCE

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- **yowo** — production YOLO11/YOLO26 inference and export across TensorRT, ONNX Runtime, CoreML and OpenVINO, with ByteTrack and CLIP-ReID tracking and hardware-aware backend selection; on PyPI. **Moon**, **Lunaris** and **ADD** ship from github.com/pilotspace through crates.io, npm and PyPI. Also **fastapi-multi-tenant** and **face-anti-spoofing**, founded at Vietmoney.

## TECHNICAL SKILLS

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**AI & LLM:** Multi-provider gateways, policy routing & failover, and cost/latency/quality tradeoffs across OpenAI, Anthropic, Gemini, Bedrock, Azure and OpenRouter; token metering & cost reconciliation; RAG and retrieval — embeddings, vector search, reranking (Weaviate, Qdrant); agent memory, MCP (client & server), prompt & agent orchestration; LLM evaluation harnesses and CI regression gates; PyTorch, YOLO, TensorRT, Triton, ONNX Runtime

**Languages & backend:** Python (typed, async), Rust, TypeScript / Node.js, Java, SQL; FastAPI, SQLAlchemy 2, Pydantic, pytest, Tokio, Next.js / React; Docker; functional & integration testing, agent-assisted delivery (frozen contracts, RED/GREEN gates, verification receipts); hexagonal architecture, CQRS, event sourcing, multi-tenancy, outbox & idempotency, PCI boundaries, OAuth / OIDC

**Data, storage & cloud:** PostgreSQL (Patroni HA, RLS, partitioning), Redis, MongoDB, DynamoDB, DuckDB; Kafka, PyFlink, PySpark, Glue, Airflow, BigQuery; Kubernetes, Helm, Envoy, OpenTelemetry, AWS, Azure, GCP

## EDUCATION, TEACHING & CERTIFICATIONS

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**Education:** BSc Computer Science (High-Quality Program), Ton Duc Thang University, 2015–2019

**Teaching & speaking:** Design Patterns instructor, Ton Duc Thang University (2022); data-analysis mentor, MindX (2023); speaker, AI Workshop, FPT University (2025)

**Certifications:** Google Cloud Technical Series — AI Agents (2025); AWS Machine Learning Foundations (2022)

**Languages:** English — the working language of all my published code, documentation and benchmark reports; Vietnamese